

FEATURE CLASSIFICATION PART 2 - ADJUSTING WITH PRINCIPAL COMPONENT ANALYSIS AND CALCULATING THE ACCURACY

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The intuition behind how we're deciding on how many PCs to keep is based on the following factors:

- 1) **How much variance does each PC represent:** After running PCA, we get a 'explained_variance_ratio_' array (under the pca() function in scikitlearn), that stores the total variance each PC explains. We then plotted a scree plot of this data of PC vs explained variance percentage, and we see that the curve starts flattening around PC number 10-11.

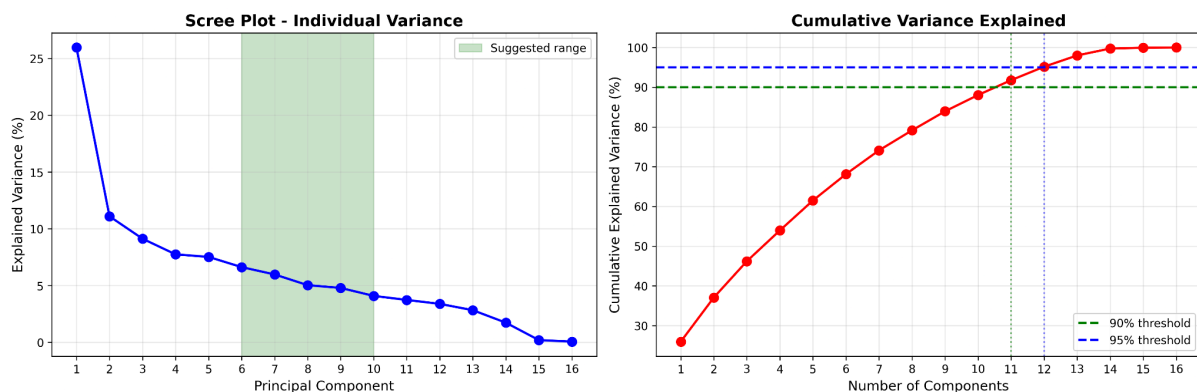


Fig. - Plots showing individual variance vs PC and Cumulative Variance vs number of components

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PCA VARIANCE ANALYSIS
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--- Individual Component Variance ---
PC 1: 25.98% | Cumulative: 25.98%
PC 2: 11.10% | Cumulative: 37.08%
PC 3:  9.13% | Cumulative: 46.21%
PC 4:  7.75% | Cumulative: 53.96%
PC 5:  7.53% | Cumulative: 61.49%
PC 6:  6.63% | Cumulative: 68.12%
PC 7:  5.98% | Cumulative: 74.11%
PC 8:  5.04% | Cumulative: 79.14%
PC 9:  4.80% | Cumulative: 83.94%
PC10: 4.09% | Cumulative: 88.03%
PC11: 3.74% | Cumulative: 91.77%
PC12: 3.39% | Cumulative: 95.16%
PC13: 2.83% | Cumulative: 98.00%
PC14: 1.73% | Cumulative: 99.73%
PC15: 0.20% | Cumulative: 99.93%
PC16: 0.07% | Cumulative: 100.00%

--- Variance Thresholds ---
Components for 90% variance: 11
Components for 95% variance: 12
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Fig. -Terminal output showing the individual and cumulative variance at each PC

- 2) **Select number of PCs that explain a cumulative variance of around 90-95%:**

The cumulative variance at every PC tells us how much overall covariance is represented by how many PCs. From the plot, we can see that around PC number 11-12, the cumulative variance reaches around 90%, which means that over 90% of variance is explained by the first 11-12 PCs.

- 3) **Calculate the accuracy of the KNN output with different number of PCs:** The logic here is to calculate how much accuracy do 4, 6, 8, 10, 12 number PCs give us. This will show us the amount of accuracy that we get for keeping 4, 6, 8, 10, 12, 16

PCs (for $k=5$ for KNN). There is a point where the accuracy will reach it's maximum or even taper off, and we would prefer to keep these PCs.

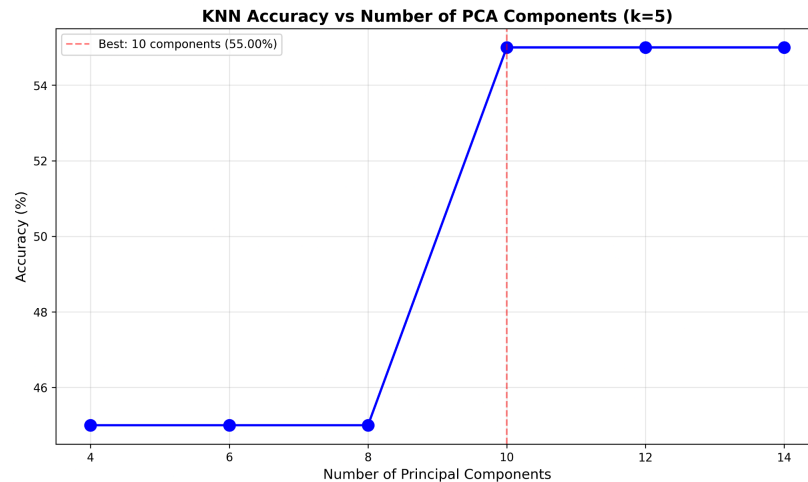


Fig. - Plot of KNN accuracy vs number of PCA components

Depending on the above three mentioned criteria, we will select a sweet spot where the accuracy is high and the cumulative variance represented is above 95%. From the KNN accuracy vs PC graph, you can see that we get an accuracy of 55% from the PC 10 and above. But if we check the cumulative variance vs PC graph, we see that PC 10 only gives us a cumulative variance of around 88%. Hence we chose the total number of PCs as 12, because for these many number of PCs, the accuracy is high (55%) and the cumulative variance that is represented is also above 95%.

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STEP 4: TRAINING FINAL MODEL WITH BEST n_components=12
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Best PCA Model:
Number of components: 12
Accuracy: 0.5500 (55.00%)
Variance explained: 95.16%

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STEP 5: FINAL COMPARISON
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Baseline (16 features): 0.6000 (60.00%)
PCA (12 components): 0.5500 (55.00%)

Improvement: -0.0500 (-5.00%)
x PCA reduced classification accuracy
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Fig. - Terminal output of comparison of baseline accuracy (with the original 16 features) vs the accuracy after PCA

The accuracy of PCA (55%) was lesser than the baseline accuracy (60%) that was calculated using just the raw z-score normalised features. The decrease in accuracy in PCA could have been because we are purely approaching the classification in a data perspective rather than looking at the tags in the data. Whereas KNN works by measuring the distance between data points that have been classified by tagging in the training phase. It performs well when points of the same class are close together and points from different classes are far apart.

In addition to this, the amount of data that we have is a very tiny data set, and PCA works by calculating a covariance matrix from your training data to find the directions of maximum variance. With this small dataset, the covariance matrix could be a poor estimate of the true covariance of your data.

Intuitively, that makes KNN more effective for this particular use case.